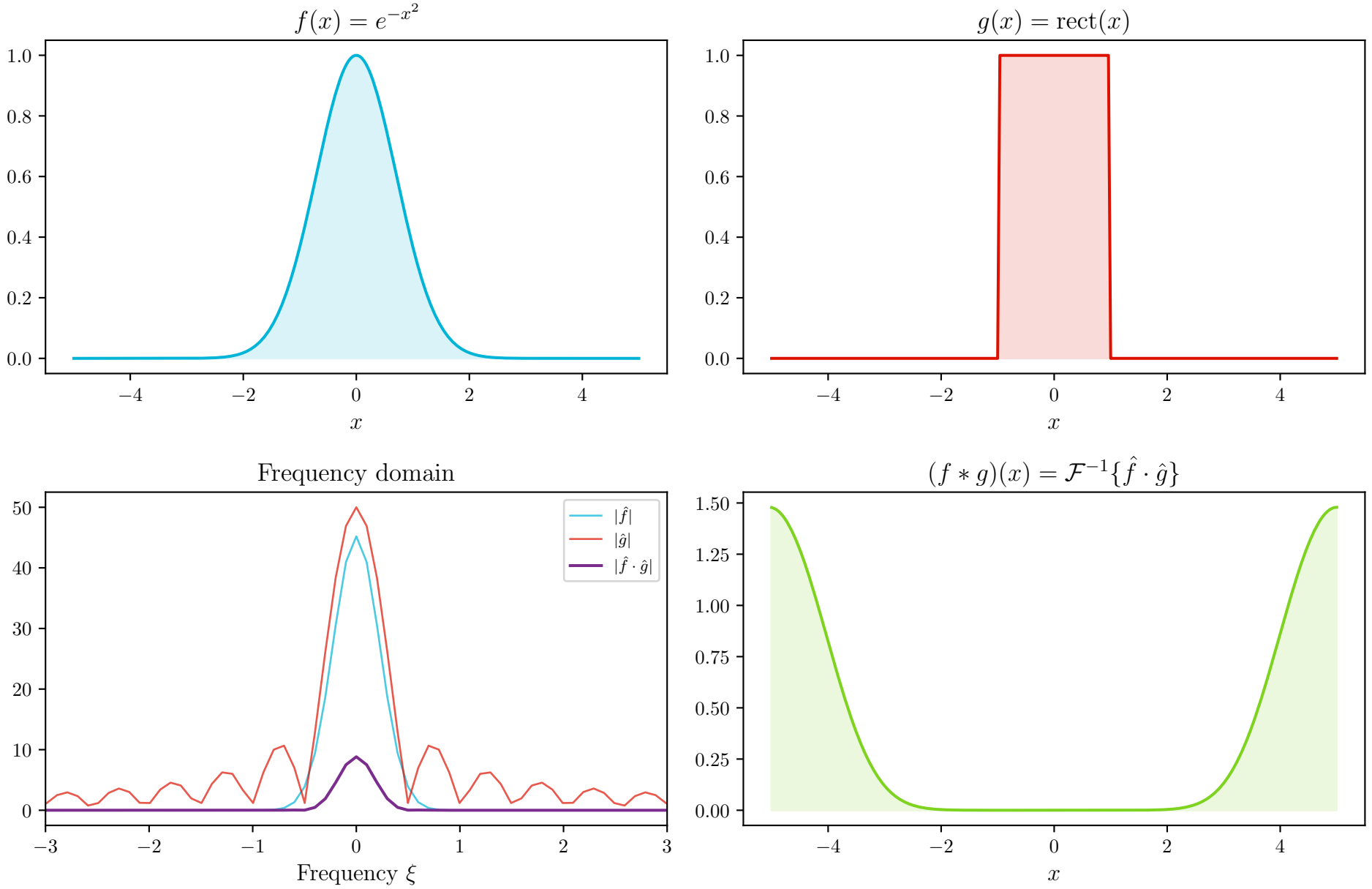
Convolution Theorem: $\mathcal{F}\{f * g\} = \hat{f} \cdot \hat{g}$
$f(x) = e^{-x^2}$
$g(x) = \mathrm{rect}(x)$
Frequency domain
$|\hat{f}|$
$|\hat{g}|$
$|\hat{f} \cdot \hat{g}|$
Frequency ξ
$(f * g)(x) = \mathcal{F}^{-1}\{\hat{f} \cdot \hat{g}\}$
x